Lab 3: Component Modelling & Architectural Pattern Selection — Written Justification

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Architecture Selection

I chose Layered Architecture for the Hostel Maintenance & Issue Ticketing System.

1. Architectural Choice

There are three tiers in the architecture of the system - that is, Layer I includes Presentation aspects which consist of the interfaces used by hostel residents, maintenance wardens, and the maintenance staff; the second layer is the Business Layer which comprises the ticket manager, notification service, and geo-location service; finally, the third layer consists of the data layer which includes the database that holds tickets and status history.

2. Two Reasons

- The ticketing process outlined by FR-001–FR-005 (log → assign → track → complete → close) is a step-by-step workflow which is directly linked with different layers of the system and allows avoiding the coordination overhead for the services that would appear in the case of the separation of the whole pipeline in the microservices architecture.
- The system operates in a single campus setting where it has a limited number of players (residents, wardens, maintenance personnel) and so microservices architecture for independent scaling is unnecessary. The usage of a layered approach allows making the codebase easier to develop and operate by a small information technology team of specialists working for the administration of the hostel.

3. Security Advantage

When all data access goes through the Ticket Manager component located in the Business Layer, this implies that the Presentation Layer avoids making any direct queries to the Database component. Introducing this concept of having a single point of control grants The Business layer the ability to enforce necessary restrictions and validation process as required. For example, it checks whether a Resident logs in before a ticket is created and so forth, which means that when any ticket, photo, or any geo-location record is transferred in or out of the Data Layer, the amount of information regarding the identity of the Resident and its location will be minimized (as per NFR-002).

4. Performance Benefit

Since the Notification Service and the Geo-Location Service are distinct Business Layer components used by the Ticket Manager and not part of it, the SLA-critical operations such as changing the status of a ticket and the timestamps (FR-003) cannot be impeded by the ow rate of the external processes such as sending the escalation alert via SMS/Email (NFR-001) or determining the geo-coordinates (NFR-002). The Ticket Manager can maintain an updated status in the Database at once while the escalation

and the geo-tagging is done via their own components.

